

Research Workshop

Bogdan G. Popescu

Course Description

This course is an intensive, hands-on introduction to writing a research article in the social sciences, with a focus on political science, sociology, or economics. Students will be taught to use R, R Studio, Quarto, Markdown, and Jabref.

By the end of the course, students will have developed and refined a complete research paper through a series of iterative assignments and structured feedback. The course integrates methodological training, substantive readings, and practical writing skills. Course content is divided into weekly units covering both technical skills and theoretical content.

This is the visual roadmap of the course:

Topic → Proposal → Methods → Analysis → Paper Draft → Feedback → Final Paper → Website

Learning Outcomes

Upon successful completion of this course the students will be able to:

- Develop a research question and transform it into a publishable paper.
- Master essential academic tools including reference managers (e.g., JabRef), markdown-based word processing, and professional presentation software.
- Understand and apply core methods in social science research, including qualitative comparative analysis, difference-in-differences (DiD), and regression discontinuity design (RDD).
- Write each section of a paper: abstract, introduction, argument, methods, literature review, findings, discussion, and conclusion.
- Conduct basic quantitative analysis in R, including data merging, regression modeling, and visualization.
- Create a personal academic website using GitHub Pages to showcase their work.

Research Paper

The research paper should provide an extensive background on the topic and a clear contribution to the literature. The analysis should include some quantitative analysis to test hypotheses. The statistical part of the research project involves using data (collect and prepare the data to run quantitative analyses and produce graphs) and specialized software (R).

Research Paper Template

Use the following as a starting point for writing your own research paper:

- [Research Paper Template](#) | [Replication](#)

Research Paper Examples

This section provides two complete examples of finished research papers and presentations. These are not templates to copy, but illustrations of different research designs. Use them to see how arguments, methods, and findings can be structured and communicated in practice.

- **Quantitative Example**
 - [Research Paper \(Quantitative\)](#) | [Replication Files](#)
 - [Presentation \(Quantitative\)](#) | [Replication Files](#)
- **Qualitative Example**
 - [Research Paper \(Qualitative\)](#) | [Replication Files](#)
 - [Presentation \(Qualitative\)](#) | [Replication Files](#)

Weekly Paper Presentations

Each week, you will give a short presentation on that week's reading. Follow [these guidelines](#)

Recommended Optional Books

Writing Academic Articles

Coppedge, M. 2012. *Democratization and Research Methods*. Cambridge University Press.

Halperin, S., & Heath, O. (2012). *Political Research: Methods and Practical Skills* (3rd ed.). Oxford University Press.

Punch, K. F. 2014. *Introduction to social research: Quantitative and qualitative approaches* (3rd ed.). SAGE Publications.

Van Evera, S. 1997. *Guide to methods for students of political science*. Cornell University Press.

Statistics

Cunningham, Scott. 2021. *Causal Inference: The Mixtape*. Yale University Press. <https://mixtape.scunning.com>.

Ismay, Chester, and Albert Y. Kim. 2019. *Statistical Inference via Data Science: A Modern Dive into R and the Tidyverse*. Chapman and Hall / CRC. <https://moderndive.com/>.

Huntington-Klein, Nick. 2021. *The Effect: An Introduction to Research Design and Causality*. Boca Raton, Florida: Chapman and Hall / CRC. <https://theeffectbook.net/>.

Llaudet, Elena and Imai, Kosuke. 2023. "Data Analysis for Social Science." Princeton: Oxford University Press.

Warne, Russel T. 2018. "Statistics for the Social Sciences. A General Linear Model Approach." London: Cambridge University Press.

Causal Inference

Bauer, Paul C. and Dennis Cohen. 2023. *Applied Causal Analysis (with R)* <https://bookdown.org/paul/applied-causal-analysis/>.

Keyes, David, R for the Rest of Us: 2025. *A Statistics-Free Introduction* <https://book.rfortherestofus.com>. No Starch Press

Geographic Information Systems (GIS)

Lovelace, Robin, Nowosad, Jakub, and Jannes Muenchow. 2021. *Geocomputation with R*. <https://bookdown.org/robinlovelace/geocompr/>.

Mieno, Tara. 2023. *R as GIS for Economists*. <https://tmieno2.github.io/R-as-GIS-for-Economists/>.

1 The Topic and Research Question

Lecture

[Download the Slides](#)

Video

[Watch the Video](#)

2 Essential Tools for Writing a Paper Slides

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

3 Operations and Objects in R Slides

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

4 Quarto Presentations

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

5 Intro to Statistics

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

6 Working with Data in R Slides

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

7 Dplyr and Basic Visualization

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

8 Visualizing Data Distributions in R

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

9 Modeling Relationships Between Variables

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

10 Interpreting Binary and Multivariate Regression Models

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

11 Differences in Differences

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

12 Regression Discontinuity Design

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

13 Data Visualization in R - 1

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

14 Data Visualization in R - 2

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

15 Data Visualization in R - 3

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

16 The Abstract and the Argument

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

17 Literature Review

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

18 Methods

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

19 Findings, Discussion, Intro, and Conclusion

Slides

[Download the Slides](#)

Video

[Watch the Video](#)

20 Making a website on GitHub: username.github.io

Slides

[Download the Slides](#)

21 Research Proposal

Slides

[Download the Slides](#)